NITESH SUBEDI

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Summary

PhD researcher developing embodied AI systems for autonomous robots, with expertise in reinforcement learning, vision-language models, and zero-shot sim-to-real transfer. Published at ICRA 2026 and experienced in building end-to-end robotic learning pipelines spanning simulation, policy optimization, perception, deployment, and evaluation on physical mobile manipulators.

Education

Iowa State University, Ames, Iowa — May 2024–Present
Ph.D. in Mechanical Engineering

Institute of Engineering, Tribhuvan University, Nepal — Nov. 2017–Apr. 2022
Bachelor's Degree in Mechanical Engineering

Professional Experience

Graduate Research Assistant – Iowa State University, Ames, Iowa — May 2024–Present

- *Find the Fruit: Zero-Shot Sim2Real RL for Occlusion-aware Plant manipulation*
  Developed a PPO-based reinforcement learning pipeline in NVIDIA Isaac Sim for occlusion-aware plant manipulation using deformable vegetation model, achieving 86.7% zero-shot success on real robots and publication at ICRA 2026.
- *Full Autonomous Harvesting Pipeline*
  Extending Find the Fruit from occlusion-aware manipulation toward a complete harvesting pipeline that integrates perception, manipulation, and task execution. In progress.
- *Language-Conditioned Latent Alignment (LCLA) for Vision language Navigation*
  Proposed Language-Conditioned Latent Alignment (LCLA), aligning frozen vision-language embeddings with a privileged navigation policy to enable zero-shot language navigation on TurtleBot platforms (~80% unseen-environment success). Released as arXiv:2602.07629. Accepted at CoRL 2026
- *Language-Conditioned Manipulation*
  Extending LCLA from navigation to manipulation tasks to study language-conditioned robot behavior across mobile and manipulation settings. In progress.
- *Agentic Manipulation and Navigation*
  Developing agentic manipulation and long-horizon navigation pipelines for Stretch 3 and Stretch 4 mobile manipulators using VLM-guided planning and autonomous task execution.

R&D Engineer – North Star Developer's Village (NSDeVil), Nepal — Dec. 2021–Apr. 2024

- Led a five-person team developing autonomous navigation for wheeled robots operating around people in classroom environments.
- Developed a custom visual SLAM approach for navigation in unknown outdoor environments.

Mechanical Engineering Intern – E-Bolt Tech Pvt. Ltd., Nepal — Apr. 2021–May 2021

- Compared two-wheeler suspension systems and built a Python model of motorcycle dynamics with ADAMS–MATLAB tire-road co-simulation.

Robotics Club – Institute of Engineering, Tribhuvan University, Nepal — Aug. 2018–Aug. 2021

- Led mechanical design and control for ABU Robocon 2020, mentored the 2021 team, and built pneumatic kicking, four-legged walking, and roller-based ball-shooting mechanisms.

Selected Publications

- **N. Subedi***, H-J Yang*, D.K. Jha, S. Sarkar. "Find the Fruit: Zero-Shot Sim2Real RL for Occlusion-Aware Plant Manipulation." *IEEE International Conference on Robotics and Automation (ICRA)*, 2026. Accepted.
- **N. Subedi**, A. Haroon, S. Tetteh, P. Koirala, C. Fleming, S. Sarkar "LCLA: Language-Conditioned Latent Alignment for Vision-Language Navigation." *arXiv:2602.07629*, 2026. Under Review.
- **N. Subedi**, A. Haroon, S. Ganguly, S. Tetteh, P. Koirala, C. Fleming, S. Sarkar "Can Pretrained Vision-Language Embeddings Alone Guide Robot Navigation?" *arXiv:2506.14507*, 2025.

Skills

Programming	Python, C, MATLAB, Git
Frameworks	PyTorch, ROS, ROS 2, CasADi, IPOPT, Numba
Robotics / Sim	NVIDIA Isaac Sim, ADAMS, TurtleBot, Stretch 3, Stretch 4, LiDAR, Arduino, Raspberry Pi
ML / Control	Reinforcement learning, vision-language models, sim-to-real transfer, MPC, DPC, autonomous navigation
Systems	Policy training and evaluation, robot deployment, containerization, embedded systems